

Assignment 4: Real batteries and voltmeters

Start Assignment

- Due Monday by 10p.m.
- Points 20
- Submitting a file upload
- File Types pdf
- Available Mar 24 at 2p.m. - Apr 10 at 10p.m.

Assignment Description

For this assignment, you essentially only need to finish what you started in tutorial, where you investigated the internal resistance of a battery and of your voltmeter. Your submission should include two figures along with a brief description of your method and discussion of your results.

Part 1 - Battery internal resistance

- [100 words max] Briefly explain your methodology (e.g. how you chose to vary the resistance: potentiometer? resistors?, over what resistance range? how did you measure the resistance value?, how did you determine your uncertainties?)
- Include a figure that shows the I-V curve for a resistance of varying value connected to a 9V battery. Fit the curve and give the internal resistance of the battery in the caption. You can include fit parameters in the plot or in the caption.
- [100 words max] Briefly comment on any marked deviation from the expected linear fit and if the obtained value of the internal resistance is reasonable.

Part 2 - Voltmeter internal resistance

- [100 words max] Briefly explain your methodology (e.g. why you picked the capacitor you picked, how you recorded the time-dependent data, how you determined your uncertainties)
- Include a figure that shows the discharge voltage vs time of a capacitor connected to your voltmeter. Fit the curve and give the time constant **and** value of the internal resistance of your voltmeter in the caption. You can include fit parameters in the plot or in the caption.
- [100 words max] Briefly comment on if the obtained value of the internal resistance is reasonable and on the implication of your results when making measurements with your voltmeter.

Submission details

Your assignment submission should be single PDF document that includes the two parts as described above.

Collaboration and Academic Integrity

For all long-form assignments, you are encouraged to collaborate with others while collecting and analyzing your data. Working together to brainstorm ideas, troubleshoot experimental setups and debug code are all expected behaviors of professional physicists. While you are free to share equipment (like battery, DMM, etc.), but the data you collect, the data analysis you perform on these data and your final written submission must be your own work. If you have more questions about this, reach out to someone on the teaching team and/or remind yourself of the course's [academic integrity guidelines \(https://canvas.ubc.ca/courses/181842/pages/academic-integrity?module_item_id=8207350\)](https://canvas.ubc.ca/courses/181842/pages/academic-integrity?module_item_id=8207350).

Assignment 4

Criteria	Ratings		Pts
Part 1 - Methods Clearly and concisely explains experimental procedure and includes key facts about methodology (e.g. how resistance was varied and measured, over what range and how uncertainties were determined).	2 to >0.0 pts Full Marks	0 pts No Marks	2 pts
Part 1 - Figure for I-V data Sufficient number of high quality data that spans an appropriate range along with a line of best fit are included in a single plot. Appropriate errorbars are included for the data points. Graph is clear and readable with appropriate line styles and contrasting colors. Axes are labeled with units and a clear legend is included. A brief and informative caption is provided that includes a description of fit performed, the fit parameters (if not on the plot) and the internal resistance obtained. Uncertainties are included where necessary.	5 to >0.0 pts Full Marks	0 pts No Marks	5 pts
Part 1 - Discussion Any marked deviation from the expected linear fit is meaningfully commented on and explained. Whether the obtained internal resistance is reasonable is discussed and supported with reference(s).	3 to >0.0 pts Full Marks	0 pts No Marks	3 pts

Criteria	Ratings		Pts
Part 2 - Methods Clearly and concisely explains experimental procedure and includes key facts about methodology (e.g. why the capacitor picked was chosen, how the time-dependent voltage was recorded, and how uncertainties were determined).	2 to >0.0 pts Full Marks	0 pts No Marks	2 pts
Part 2 - Figure for discharge voltage Sufficient number of high quality data that spans an appropriate range along with a line of best fit are included in a single plot. Appropriate errorbars are included for the data points. Graph is clear and readable with appropriate line styles and contrasting colors. Axes are labeled with units and a clear legend is included. A brief and informative caption is provided that includes a description of fit performed, the fit parameters (if not on the plot) and the internal resistance obtained. Uncertainties are included where necessary.	5 to >0.0 pts Full Marks	0 pts No Marks	5 pts
Part 2 - Discussion Whether the obtained internal resistance is reasonable is discussed and supported with reference(s). Significance of a finite internal resistance is meaningfully discussed in the context of making measurement with the voltmeter.	3 to >0.0 pts Full Marks	0 pts No Marks	3 pts
Total Points: 20			